

ENGR 2303 Electrical Science
Review Material for Test 3 (7/14/26)

The test will cover circuit analysis of first order RC circuit and RL circuit.

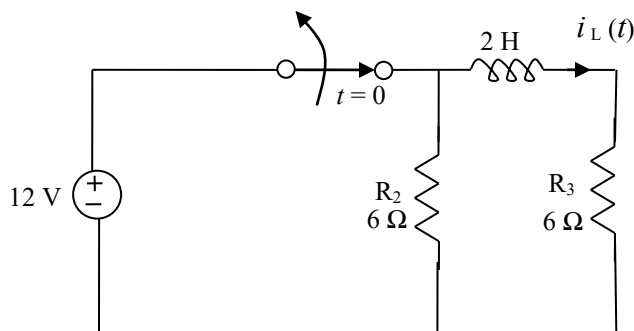
You should be able to do the following

1. Find the voltage across the capacitor if current is given.
2. Find the current through the inductors if voltage is given.
3. Add capacitors or inductors in parallel and in series
4. Use voltage and current divider rules for capacitors and inductors.
5. Find the energy stored in capacitors and inductors when there is DC voltage or current in the circuit.
6. Find the voltage of the capacitor in source-free circuit
7. Find the current through the inductor in source-free circuit.
8. Find the voltage of the capacitor when there is a DC voltage or current source.
9. Find the current in the inductor when there is a DC voltage or current source.

Sample Questions

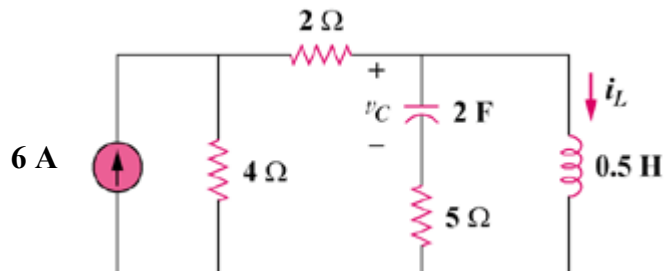
Question 1

The switch was closed for a long time and at $t=0$ the switch was open. Find $i_L(t)$ for $t \geq 0$. What is the current value when $t = 2\tau$? τ is the time constant.



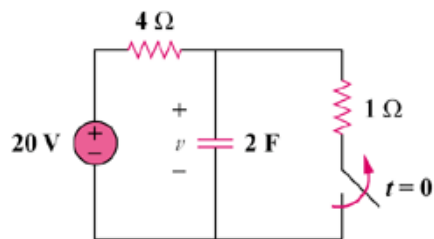
Question 2

Find v_C , i_L , and the energy stored in the capacitor and inductor in the circuit under dc, steady-state, conditions.



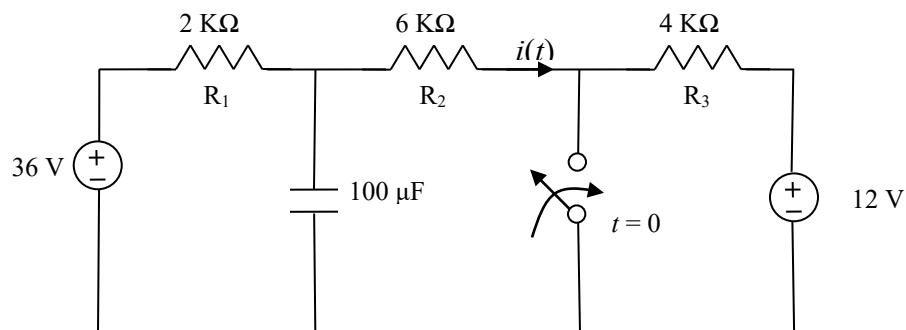
Question 3

Calculate the capacitor voltage for $t < 0$ and $t > 0$ for circuit below.



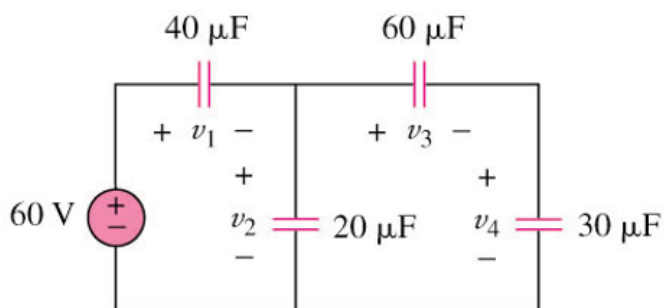
Question 4

Use the differential equation approach to find $i(t)$ for $t > 0$. The circuit is in steady state prior to time $t = 0$, when the switch is closed.



Question 5

Find the voltage across each of the capacitors in the circuit shown below:



Question 6

The circuit below is used by a biomedical engineering student to study the "frog kick." She noticed that the frog kicked a little when the switch was closed but kicked violently for 5 second when the switch was opened. Model the frog as a resistor and calculate its resistance. Assume that it takes 10 mA for the frog to kick violently.

